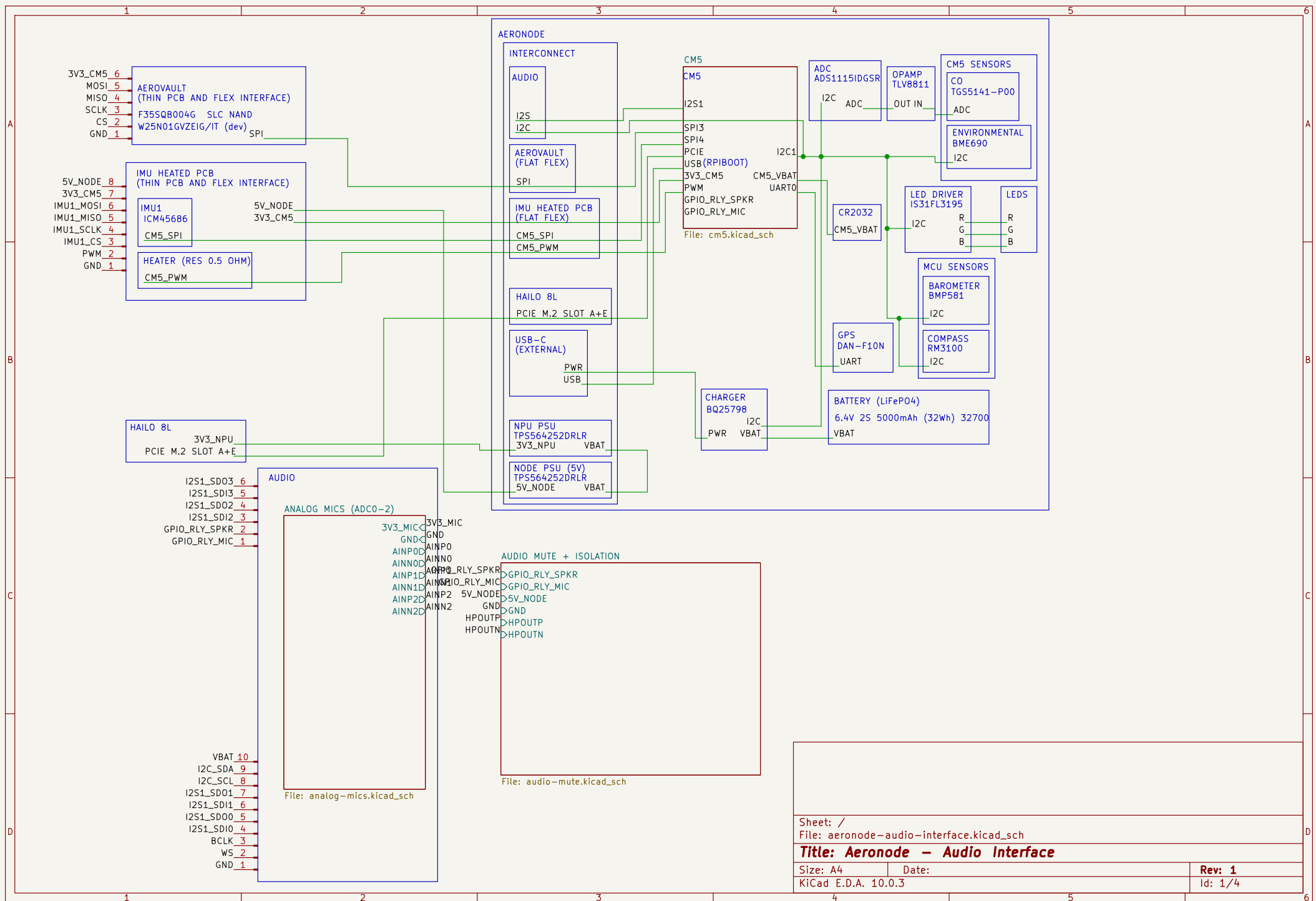
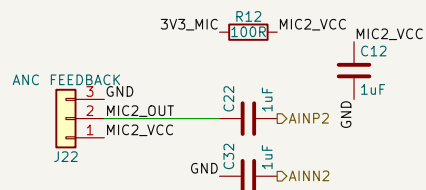
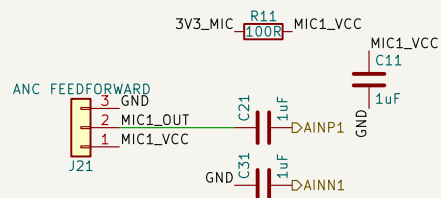
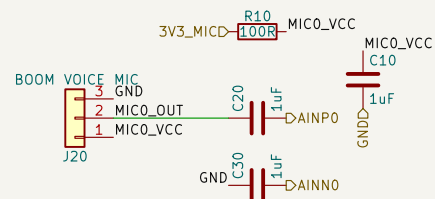




1	2	3	4	5	6
A					A
B					B
C					C
D					D
1	2	3	4	5	6

Sheet: /CM5/ File: cm5.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 2/4



ANALOG MICS -> ADAU1860 ADC0/1/2 LIMA 2026-09-09 lima:docs/h1-headset-mute-relay.md

CHANNEL BUDGET IS NOW FULL. The ADAU1860 has exactly THREE ADCs (AINP/N 0,1,2; datasheet: "three ADC channels and one DAC channel"). Three mics uses all of them, ZERO spare. Nothing else analog can be added without changing part.

ALL THREE MICS ARE ON THE AERONODE SIDE. They are our mics, biased from our rail and referenced to our GND, so they never touch AC_GND. The isolation boundary on audio-mute.kicad_sch is untouched - no transformer is needed here.

FRONT END, per channel: Rn (100R) + Cn (1uF) filter the supply at the connector. C2n / C3n couple OUT and the mic's local GND into AINPn / AINNn – pseudo-differential, so cable ground noise is rejected. This mirrors the EVB, which puts 22uF in series with BOTH legs and no bias network (UG-2017 Figure 8).

USE AMPLIFIED MIC MODULES, NOT BARE CAPSULES.
The ADAU1860 has NO MICBIAS pin, and its PGA is only 0-24 dB against a 0.49Vrms single-ended full scale. A bare electret gives a few mV: x16 lands about -24 dBFS at best and the rest has to come from digital gain, which lifts the noise floor with it. Modules must output a few hundred mV and run off 3V3_MIC.

3V3_MIC MUST BE ITS OWN QUIET RAIL – not the CM5's 3V3. ANC noise performance is set by the mic supply. [gap] No LDO is specified yet, which is why 3V3_MIC shows one deliberate "label not connected" ERC error on the parent sheet.

ANC IS ALWAYS ON (ruled by Peter, 2026-09-09)
K1 no longer switches AeroNode in and out - AERONODE_AUDIO is permanently summed into HS_L/HS_R through R1/R2 (220R) on audio-mute.kicad_sch, so the DAC always reaches the earcup. K1 breaks only the intercom feed.

[gap] THE OPEN QUESTION IS ANC AUTHORITY, NOT CONNECTIVITY. A passive sum cannot fight a low-impedance source: while the intercom is live, AeroNode sits about -33 dB at the earphone (-7.6 dB once K1 opens the intercom). So ANC in NORMAL flight depends on the panel's headphone output impedance, unmeasured.

If the panel is stiff, ANC will need its own path to the transducer rather than sharing the intercom line. Section 21 has the numbers. MEASURE THE PANEL.

LATENCY: at 48 kHz the ADC->DSP->DAC loop is ~1 ms, too slow for feedforward above a few hundred Hz. Run ANC through FastDSP at a high rate – the datasheet characterises it at 768 kHz for exactly this.

T1 is not a bandwidth limit here: its 200 Hz–4 kHz spec is Bourns' 600R telecom condition; into 160R from a low-Z drive the corner computes to 3.4 Hz.

Sheet: /ANALOG MICS (ADC0-2)/
File: analog-mics.kicad_sch

Title:

Size: A4

Date:

Rev:

KiCad E.D.A. 10.0.3

Id: 4/4